Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Fluidodinamica Computazionale
Dipartimento di Scienze e Tecnologie Aerospaziali
Politecnico di Milano

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Chapter 3

Summary

[*Briefly describe the goal of the project.*]

Chapter 4

Problem definition and background

[*Introduce the problem and state the objective of your work. Briefly present the state of the art regarding the chosen topic and report a reference solution (i.e. numerical or experimental, or the exact one if available).*]

4.1 Literature review

4.2 Reference solution

Chapter 5

Design of Experiment

[*Describe the process used to meet the project goal.*]

Chapter 6

Computational model

[*Describe thoroughly the computational model/s used in the project*]

6.1 Problem geometry and setup

6.2 Mesh generation and description

6.3 Numerical schemes

Chapter 7

Results

[*Report the results of the simulations. Validate your work, i.e. show that the computational model (6) and the simulations you run (the DoE 5) were able to obtain the goal of the project*]

7.1 Test 1

7.1.1 Grid convergence

Chapter 8

Conclusions

Appendix A: Resources

[*Report the config files of the software used (i.e. SU2 [**?**] and the mesher). Also attach to this report an archive with the mesh files, solutions and the reference solution data (e.g. data points of a Cp plot ...)*]

Mesh configuration files

SU2 configuration files